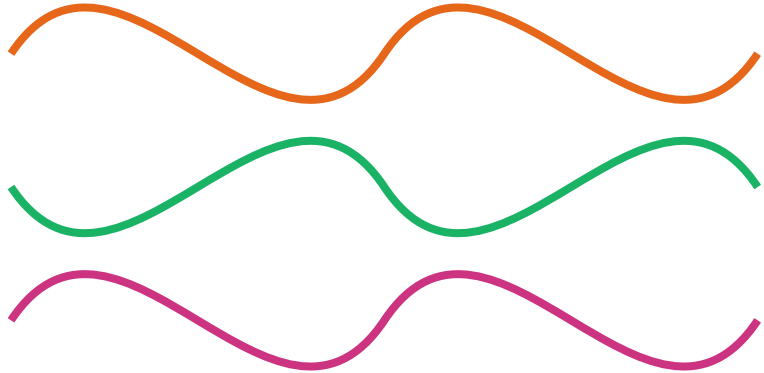


Page 1: Native Vector Pipeline

Rendered from the vector display list: zoom re-rasterizes vectors, never a bitmap



Vector Kerning Test

Glyph positions come from PDF font metrics, not UI text layout

ENGINE STATUS: 100% Native Vector

- o In Auto mode: Status bar displays lightning badge 'Vector'.
- o Visual sharpness: Zoom into this page up to 500% - lines and curves stay pin-sharp!
- o No bitmap scaling: every zoom level replays the same display list.

Page 2: Hybrid Fallback Pipeline

Region fallback: only the unsupported objects are composited from PDFium

NOTICE: The two panels below form a knockout transparency group

The vector engine classifies them as a TransparencyGroup fallback region and composites just those regions from PDFium; the rest stays vector.

Rendered via PDFium
100% Visual Fidelity

Zero Truncation
Safe Content Delivery

HOW TO VERIFY FALLBACK DIAGNOSTICS:

1. Look at the bottom-right status bar: it displays 'Hybrid (N PDFium regions)'.
2. Hover your mouse over the badge: the tooltip lists the reason:
'TransparencyGroup'.
3. Switch to 'PDFium Only' in View -> Rendering Engine: renders via standard PDFium.
4. Switch to 'Vector Only' in View -> Rendering Engine: the regions are outlined.
5. Return to 'Auto': transparent fallback is restored instantly.

Page 3: Real-Time Engine Switching

Compare the vector engine and PDFium text rendering at all scales

7pt: The quick brown fox jumps over the lazy dog 1234567890 - Subpixel Font Hinting

9pt: The quick brown fox jumps over the lazy dog 1234567890 - Subpixel Font Hinting

11pt: The quick brown fox jumps over the lazy dog 1234567890 - Subpixel Font Hinting

14pt: WPF glyph runs vs PDFium software rasterizer

20pt: Retained vector drawing

28pt: High-DPI Scalability



INTERACTIVE TEST EXPERIMENT:

1. Open the top menu: View -> Rendering Engine.
2. Switch between 'Auto' (Vector) and 'PDFium Only' (Raster).
3. Observe how the page instantly re-renders in real time.
4. Notice subtle differences in font smoothing and stroke weights.
5. Check the status bar: the badge updates dynamically to match the active engine!